Nutritional Assessment in Pediatric Gastroenterology: Z-Scores, Mid-Upper Arm Circumference, the ASPEN and Academy Criteria, Bone Age, and Bone Density Adjusted for Height.

Picture a nine-month-old boy with biliary atresia. His weight is on the twentieth centile, a z-score of minus zero point eight, and the clinic letter says that his growth is satisfactory. But his abdomen is tense with ascites, his liver and spleen are large, and his arms are thin. When the dietitian measures the circumference of his upper arm, it is more than two standard deviations below the mean for his age. Most of the weight on the scale is fluid and organ, not fat and muscle.

Now picture a thirteen-year-old girl with Crohn's disease. Her body mass index is a little low, but not dramatically so. What has changed is her height: a year ago her height z-score was minus zero point four, and today it is minus one point two. She has not started puberty. A dual-energy X-ray absorptiometry scan, the bone density test usually called DXA, reports a lumbar spine z-score of minus two point four, and someone has written osteoporosis in the problem list.

The clinical question of this chapter is how to decide whether a child in a gastroenterology clinic is malnourished, how badly, and what that means for the skeleton. The main idea is that nutritional assessment is a measurement problem. You need the correct growth chart, you need to think in z-scores rather than percentiles, you need a trajectory rather than a single point, you need an arm measurement wherever weight is misleading, and you need to read the skeleton in the light of the child's height. Malnutrition is then written as a diagnosis with a grade, a cause and a time course, and it gets a follow-up plan like any other diagnosis.

A note on sources. For growth charts we use the 2010 recommendation of the United States Centers for Disease Control and Prevention, endorsed by the American Academy of Pediatrics and written by Grummer-Strawn and colleagues, and the 2007 World Health Organization reference for school-age children by de Onis and colleagues. For the definition and grading of malnutrition we use the consensus statement of the American Society for Parenteral and Enteral Nutrition and the Academy of Nutrition and Dietetics, published by Becker and colleagues in 2014, which we will call the ASPEN and Academy criteria, together with the paper by Mehta and colleagues in 2013 that set out the new definition, and a later review by Bouma in 2017. For the European view we use the 2022 position statement of the special interest group on clinical malnutrition of ESPGHAN, the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, by Hulst and colleagues. For the burden of malnutrition and screening tools we use two large European multicenter studies, by Hecht and colleagues in 2015 and Chourdakis and colleagues in 2016. For arm circumference we use the mortality study by Mramba and colleagues in the British Medical Journal in 2017 and a study of children with cholestasis by Silva and colleagues in 2016. For laboratory markers we use the 2021 ASPEN position paper on albumin and prealbumin by Evans and colleagues. And for bone we use the 2019 pediatric positions of the International Society for Clinical Densitometry and the height adjustment work of Zemel and colleagues.

Part one. Choosing the chart and thinking in z-scores.

Two kinds of growth chart exist, and the difference matters. A standard describes how children should grow under good conditions. The World Health Organization standards of 2006, for children from birth to five years, were built from healthy breastfed infants in six countries, all of them breastfed for the first twelve months. A reference describes how a population actually did grow. The Centers for Disease Control and Prevention charts of 2000, for ages two to nineteen, describe a sample of American children of whom only about half were ever breastfed. The 2010 recommendation therefore says to use the World Health Organization standards for every child under twenty-four months, whatever the feeding, and the Centers for Disease Control and Prevention charts from two to nineteen years, switching at twenty-four months, which is also when we move from measuring length lying down to measuring height standing up. Outside the United States, the World Health Organization 2007 reference covers ages five to nineteen; it is constructed so that, at age nineteen, a body mass index one standard deviation above the median matches the adult overweight cut-off of twenty-five and two standard deviations matches the adult obesity cut-off of thirty. In 2022 the Centers for Disease Control and Prevention added extended body mass index charts that reach the ninety-ninth and higher centiles, defining severe obesity as a body mass index at least one hundred twenty percent of the ninety-fifth centile or at least thirty-five.

Why does the chart matter? Because the chart can create or erase malnutrition. Using the older American reference, between seven and eleven percent of American children aged six to twenty-three months are labelled low weight-for-age; using the World Health Organization standard, fewer than three percent are.

Next, think in z-scores. A z-score tells you how many standard deviations a measurement lies above or below the median for age and sex. A z-score of minus two sits at about the two point three centile, which the World Health Organization charts label as the second centile, while the Centers for Disease Control and Prevention charts traditionally flag the fifth centile. The great advantage of a z-score appears below the last printed centile. A percentile stops changing once a child falls off the chart, while a z-score keeps falling and shows the depth of the deficit: a child at minus three is in a very different place from a child at minus two, although both are below the first centile.

Most important of all, the change matters more than the point. A child who has always tracked along the third centile is usually healthy. A child who has fallen from the seventy-fifth centile to the twenty-fifth is not, even though the twenty-fifth centile is normal. The 2022 ESPGHAN statement says that an unintentional fall of one z-score or more should prompt a full nutritional assessment. Height must also be read against the parents. The target height is the average of the parents' heights, plus six and a half centimeters for a boy or minus six and a half for a girl, which is the same as adding or subtracting thirteen centimeters before dividing by two, and the normal range is about eight and a half centimeters on either side. A height z-score of minus one and a half is reassuring when both parents are short and alarming when both are tall.

Part two. Why it matters and how it happens.

Malnutrition in hospitalized children is common and costly. Hecht and colleagues studied two thousand five hundred sixty-seven children aged one month to eighteen years in fourteen hospitals across twelve European countries. Seven percent had a body mass index more than two standard deviations below the mean on admission, with a range between centers of four to about nine percent, and the figure was highest in infants, at nearly eleven percent. Moderate malnutrition added about one point three days to the hospital stay, and severe malnutrition added one point six days. And those figures are probably underestimates for gastroenterology patients, because body mass index misses children whose weight is inflated by edema, ascites or large organs.

It helps to understand the order in which undernutrition appears in the measurements. It begins with a deficit: intake, absorption or utilization of nutrients falls short of needs, and inflammation both raises needs and suppresses appetite. The first measurable consequence is a slower rate of weight gain, which in infants is the earliest sign of all. Next comes wasting, a fall in weight-for-height and in arm circumference, as the body uses up its fat and then its muscle. If the deficit lasts for months, linear growth slows and height-for-age falls; this is stunting, the signature of chronic malnutrition. Finally the skeleton records the history: bone age and puberty are delayed and bone mass lags behind. Delay is not all bad news, because open growth plates mean that some height can still be recovered once the cause is treated.

The ASPEN and Academy criteria call malnutrition acute when it has lasted less than three months and chronic when it has lasted longer, and they separate illness-related malnutrition, caused by an acute event such as trauma, burns or infection or by a chronic disease, from malnutrition that is not related to illness, which has environmental or behavioral causes. Crohn's disease deserves a special note. Inflammatory cytokines suppress the growth hormone and insulin-like growth factor axis directly, so height can fall before weight does. In a child with Crohn's disease, slow linear growth is a marker of disease activity, not just of poor intake.

Part three. Grading malnutrition with the ASPEN and Academy criteria.

The consensus applies to children from one month to eighteen years of age. It offers two sets of indicators, depending on whether you have one measurement or several.

When you see a child only once, there are four primary indicators: weight-for-height z-score, body mass index-for-age z-score, length or height-for-age z-score, and mid-upper arm circumference z-score. For weight-for-height, body mass index and arm circumference, a z-score between minus one and minus one point nine is mild malnutrition, between minus two and minus two point nine is moderate, and minus three or below is severe. Height-for-age is treated differently. The workgroup found insufficient data to grade it at minus two, so only a height z-score of minus three or below counts, and it counts as severe. This is a real point of difference, because the World Health Organization defines stunting at minus two, and Bouma, reviewing the criteria, recommended adding a moderate grade. Arm circumference z-scores are recommended between six months and six years of age, which is the range covered by the World Health Organization arm reference.

When you have two or more measurements over time, four more indicators become available. For children under two years, weight gain velocity below seventy-five percent of the expected rate is

mild malnutrition, below fifty percent is moderate, and below twenty-five percent is severe. For children aged two to twenty years, weight loss of five percent of usual body weight is mild, seven and a half percent is moderate, and ten percent is severe. A decline in weight-for-length or weight-for-height z-score of one standard deviation is mild, two is moderate and three is severe. And an intake of fifty-one to seventy-five percent of estimated energy or protein needs is mild, twenty-six to fifty percent is moderate, and twenty-five percent or less is severe.

Two rules make the system work. First, unlike the adult criteria, which need two characteristics, a single pediatric indicator is enough to make the diagnosis. Second, the most severe indicator sets the grade. Notice also that inadequate intake is an indicator in its own right. The consensus points out that in a child at risk, a nutritional deficit can begin within two or three days of poor intake, long before the scale shows anything.

Part four. The arm circumference.

The mid-upper arm circumference, measured at the midpoint between the tip of the shoulder and the point of the elbow on a relaxed left arm, deserves far more use than it gets in gastroenterology clinics. The reason is simple. The upper arm contains no organs and little fluid, so it measures fat and muscle directly. Steroids, edema, an enlarged liver and spleen, and ascites all inflate body weight, but none of them inflates the arm.

The arm is also a powerful predictor of outcome. Mramba and colleagues built an arm circumference reference for children aged five to nineteen and tested it in a Kenyan cohort against death within one year. The area under the receiver operating characteristic curve, a measure of how well a test separates those who will die from those who will not, where one is perfect and one half is a coin toss, was zero point seven three for arm circumference z-score and only zero point five eight for body mass index z-score, a highly significant difference. Children with an arm z-score below minus three had about five times the risk of death. The measurement is also reliable: in infants under six months measured by community health workers, the agreement between repeated arm measurements, expressed as an intraclass correlation coefficient, was zero point nine six, compared with zero point seven one for weight-for-length. And for children from six months to five years, the World Health Organization cut-off for severe acute malnutrition is an arm circumference below one hundred fifteen millimeters, raised from one hundred ten millimeters in 2009.

For our patients the most striking data come from liver disease. Silva and colleagues in Brazil assessed ninety-one children with cholestasis. Weight-for-height below minus two z-scores identified twelve percent as undernourished, and body mass index identified sixteen percent. Arm circumference identified forty-four percent, more than three and a half times as many. In the children older than one year, arm muscle circumference, which subtracts the fat layer estimated from the triceps skinfold, was depleted in seventy-one percent. Their conclusion, which echoes the classic 1990 paper by Sokol and Stall on anthropometry in children with chronic liver disease, is that weight-based indices underestimate malnutrition in liver disease.

A measurement needs a reference to become a z-score. Beyond the World Health Organization reference for three to sixty months, there are now American references from national health survey data: one by Abdel-Rahman and colleagues covering two months to eighteen years, based on almost twenty-nine thousand children, and one by Addo and colleagues covering one to twenty years in about thirty-three thousand children, which also provides arm muscle and fat areas adjusted for height. The Mramba reference covers five to nineteen years, and the older Frisancho norms from 1981 provide triceps skinfold and arm areas from one to seventy-four years. The practical message is to measure the arm at every visit, record both the value and its z-score, and grade malnutrition on the arm whenever weight is unreliable.

Part five. Screening tools.

Hospitals screen children on admission to decide who needs a full dietetic assessment. The commonest tools are the Paediatric Yorkhill Malnutrition Score, the Screening Tool for the Assessment of Malnutrition in Paediatrics, and the Screening Tool for Risk on Nutritional Status and Growth, known as STRONGkids. Chourdakis and colleagues applied all three to the same two thousand five hundred sixty-seven European children. The Yorkhill score rated twenty-five percent as high risk, the Screening Tool for the Assessment of Malnutrition twenty-three percent, and STRONGkids only ten percent. The three tools agreed on only forty-one percent of children, although for every tool, high-risk children stayed about one and a half to almost two days longer. In its original validation, the nurse-rated Yorkhill score had a sensitivity of fifty-nine percent against a full dietetic assessment, and a simpler Australian tool, the Paediatric Nutrition Screening Tool, had a sensitivity of seventy-eight percent and a specificity of eighty-two percent. The lesson is that screening tools are a net to catch children for assessment, not a substitute for measuring them. The 2022 ESPGHAN statement asks, as a minimum, that every child have weight and height measured, and that every facility have a pathway from screening to assessment with access to a dietitian.

Part six. What the blood can and cannot tell you.

Albumin and prealbumin are not nutritional markers. The 2021 ASPEN position paper by Evans and colleagues explains why. During inflammation the liver reprioritizes its protein synthesis toward acute phase proteins, and capillary leak moves albumin out of the circulation. Low albumin therefore reflects inflammation and predicts poor outcomes, but it should not be used as a proxy for body protein, and a normal albumin does not mean that a child is well nourished.

Micronutrients matter more. In pediatric inflammatory bowel disease, a 2025 review by Galeazzi and colleagues reported iron depletion in seventy-seven to ninety percent of children at diagnosis. Because ferritin is itself an acute phase protein, iron deficiency is defined by a ferritin below thirty micrograms per liter in remission but may be present with a ferritin up to one hundred when there is inflammation. Zinc deficiency was reported in seventeen to fifty-one percent, worst at diagnosis, in Crohn's disease and with high-output stomas. Vitamin D deficiency was reported in nineteen to forty-three percent, and ESPGHAN advises supplementing below twenty nanograms per milliliter. Vitamin B12 deficiency should be expected with terminal ileal disease or after resection of more

than about thirty centimeters of ileum, but it is uncommon at pediatric diagnosis. A word of caution about vitamin D thresholds: the 2024 Endocrine Society guideline abandoned its previous target of thirty nanograms per milliliter and all cut-offs, but it was written for generally healthy people and does not govern children with malabsorption, inflammatory bowel disease or cholestasis, in whom measuring and treating remains standard practice.

Part seven. Body composition and energy needs.

Weight is a sum; body composition is the detail. Arm circumference and triceps skinfold are bedside, cheap and resistant to edema, although skinfolds need trained observers. Bioelectrical impedance is quick and portable, but, as Kyle and colleagues emphasized in 2015, it assumes normal hydration and needs equations specific to age and ethnicity, which limits it in sick children. DXA can measure fat, lean and bone mass with regional detail and low radiation, although hydration affects the lean mass estimate. For energy needs, the ASPEN and Society of Critical Care Medicine guideline for critically ill children recommends measuring energy expenditure by indirect calorimetry where feasible and using predictive equations, such as those of Schofield, with caution, because they can under- or overestimate needs in illness. For catch-up growth, energy is commonly estimated from the needs of a child of ideal weight-for-height, divided by the child's actual weight, and the estimate is recalculated as the child gains. In a severely depleted child, watch phosphate as feeding is increased.

Part eight. Bone age.

A hand and wrist radiograph read for bone age tells you how mature the skeleton is and therefore how much growth is left. Two classic methods exist: the Greulich and Pyle atlas, which matches the film to a set of standard images, and the Tanner and Whitehouse method, which scores individual bones. They are not interchangeable. In a comparison of three hundred sixty-two radiographs, the repeatability limits for a single observer were about plus or minus two point three years with Greulich and Pyle and plus or minus one point four years with the second Tanner and Whitehouse method, and the two methods disagreed by up to about two years in either direction. Automated reading with the BoneXpert software has a precision of about zero point one seven years and agrees with the Greulich and Pyle atlas to within about zero point four years. The practical rule is to use one method, ideally automated, for serial films in the same child.

Bone age is particularly informative in inflammatory bowel disease. Gupta and colleagues studied forty-nine children with Crohn's disease with a mean chronological age of thirteen point one years and a mean bone age of twelve point two years. Forty-one percent had a bone age z-score below minus two, and delay was associated with female sex, steroids, colonic disease and thiopurine use. They recommended measuring bone age routinely. A delayed bone age is both a warning and an opportunity: it shows the cost of the disease, and it tells you that the growth plates are still open and that treating the disease can still recover height.

Part nine. DXA in children.

A child's DXA is not an adult's DXA, and the 2019 positions of the International Society for Clinical Densitometry set out the rules. The preferred sites are the lumbar spine and the total body less head; the spine can be measured from infancy and the whole body from about three years, and the hip, the forearm and the lateral distal femur are used in special situations. Results are reported as z-scores against age and sex, never as T-scores, which compare with young adult peak bone mass, and the word osteopenia is not used. A z-score of minus two or below is called low bone mineral content or density for age. Osteoporosis in a child can be diagnosed in two ways: by one or more vertebral compression fractures without local disease or high-energy trauma, whatever the bone density; or by a z-score of minus two or below together with a clinically significant fracture history, defined as two or more long bone fractures by age ten or three or more by age nineteen. Osteoporosis is never diagnosed from the densitometry alone.

The most important correction is for height. DXA projects a three-dimensional bone onto a two-dimensional image and reports areal bone density, mineral per square centimeter. A larger bone has more depth behind each square centimeter, so areal density rises with bone size even when the true density is the same. A short child has smaller bones, and compared with age-matched children of normal height, the z-score is biased downward. Zemel and colleagues used the Bone Mineral Density in Childhood Study, one thousand five hundred forty-six children plus six hundred fifty for validation, to compare correction methods, and found that adjusting for height-for-age z-score was the least biased approach. The 2019 positions therefore recommend adjusting the spine for height z-score or using bone mineral apparent density, and adjusting the total body less head for height z-score. Reference curves from about two thousand children aged five to twenty, given separately for Black and non-Black children, are available, with an online calculator for the adjustment.

This changes real decisions. In thirty-seven Turkish children with celiac disease, low bone density was much more common when judged by chronological age than when judged by height age, and density rose significantly in the first year on a gluten-free diet. In an Australian cohort of two hundred thirty-nine children with inflammatory bowel disease, only thirty percent had a DXA scan; fourteen of seventy-two scanned children had a spine z-score of minus two or below, and the height-adjusted value was usually not recorded. Many low results in short children move into the normal range after adjustment, and labelling such a child as osteoporotic can lead to the wrong drug.

Part ten. Putting it together, and what the guidelines say.

A practical assessment takes about ten minutes. Measure weight, length or height with a proper stadiometer, head circumference under two years, and arm circumference, and ask for the parents' heights. Plot on the correct chart and record z-scores, not just centiles. Look at the trajectory: weight velocity, change in z-score since the last visit, and height velocity over six to twelve months. Estimate intake as a percentage of needs and decide whether the problem is illness-related or not, and acute or chronic. If the problem is chronic or the child is short, add bone age, puberty staging and a height-adjusted DXA. Then write the diagnosis in full, for example moderate chronic illness-related malnutrition due to Crohn's disease, with a weight-for-height z-score of minus two

point three, a decline of one point two z-scores over four months and an arm z-score of minus two point one, and act on it with a dietitian, an energy target, micronutrient replacement and a date for the next measurement.

How do the documents compare? North America and Europe are aligned on using a decline in z-score: ASPEN and the Academy grade severity by a fall of one, two or three standard deviations, and ESPGHAN says that a fall of one or more should prompt assessment. They differ on the single cut-off: ASPEN and the Academy grade at minus one, minus two and minus three, whereas ESPGHAN uses a z-score below minus two as the common criterion. They differ on charts after age two, with the Centers for Disease Control and Prevention charts in the United States and the World Health Organization 2007 reference elsewhere. They are aligned in rejecting albumin as a nutritional marker, and aligned on DXA, because both follow the international densitometry positions. There is a gap in that no pediatric version of the Global Leadership Initiative on Malnutrition criteria exists; the 2025 five-year update of those adult criteria does not list one. The material difference concerns stunting. A child with a height-for-age z-score of minus two and a half and a normal weight-for-height is stunted by the World Health Organization definition but has no malnutrition diagnosis on a single ASPEN and Academy measurement. In a gastroenterology clinic, that child should be treated as chronically malnourished until the trajectory shows otherwise.

Part eleven. Traps, pearls and what to do differently.

The common traps follow from everything above. The first is weighing the ascites: a normal weight and body mass index in a child with ascites, organomegaly or steroid treatment, when the arm tells a different story. The second is treating albumin as proof, in either direction: low albumin means inflammation, and normal albumin does not mean good nutrition. The third is the unadjusted DXA: a short child with a spine z-score of minus two point four may have a height-adjusted z-score near minus one, and the wrong label leads to the wrong treatment. The fourth is the wrong chart, such as using the American reference under two years, which overcalls underweight, or relying on percentiles that hide the depth of a deficit. And the fifth is the single visit: a normal z-score that has fallen by one standard deviation since the last visit is already a diagnosis, and only a growth chart with every previous point will show it.

The numbers worth carrying are these. A z-score of minus two is the two point three centile. In the ASPEN and Academy criteria one indicator is enough, mild is minus one to minus one point nine, moderate minus two to minus two point nine, and severe minus three or below, with height counted only at minus three. Weight velocity below seventy-five, fifty and twenty-five percent, weight loss of five, seven and a half and ten percent, a z-score decline of one, two and three, and intake of fifty-one to seventy-five, twenty-six to fifty and twenty-five percent or less define the three grades. Seven percent of European inpatients have a body mass index below minus two. Arm circumference predicts mortality better than body mass index and, in cholestasis, finds more than three times as many undernourished children as weight-for-height. Screening tools agree only forty-one percent of the time. Forty-one percent of children with Crohn's disease in one series had a markedly delayed bone age. And a DXA z-score of minus two or below is low only after

adjustment for height.

So what should you do differently in clinic tomorrow? Plot every child on the correct chart and write the z-scores in the note, not only the centiles. Compare every measurement with the previous one and act on a fall of one z-score. Measure the arm in every child with liver disease, ascites, edema, steroids or a large abdomen, and grade malnutrition on the arm when weight cannot be trusted. Write malnutrition as a formal diagnosis with its grade, cause and duration. Stop ordering albumin to judge nutrition, and check iron, zinc, vitamin D and, after ileal disease, vitamin B12 instead. Order a bone age when a child with chronic disease is short or late in puberty. And never accept a DXA report on a short child until the z-score has been adjusted for height.